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Q1. Overfitting is a high-variance situation: the network has effectively memorised its training set (noise and all), so its error on fresh, held-out examples is much worse than on the examples it trained on. Remedy one: weight decay, which shrinks the parameters each step and so limits how wiggly the learned function can be. Remedy two: pruning / using a smaller architecture, which removes capacity the model would otherwise spend memorising.

Q2. Picture every word asking a question (its query) of every other word's label (its key); the match strength is the inner product of the two vectors, which come from multiplying the word embedding by three trained matrices. Normalising those match strengths with a softmax gives non-negative mixing proportions that total one, and each word's new representation is that mixture of the other words' content vectors (values). The division by the square root of the key dimension is there because inner products of long random vectors grow with the dimension; big inputs drive softmax towards a one-hot output whose gradient is almost zero.

Q3. Of the 50 emails flagged as spam, 40 really were, so precision is 0.8. Of the 60 real spam emails, 40 were caught, so recall is 40/60 = 0.667. The harmonic mean of the two is 2 x 0.8 x 0.667 / 1.467 = 0.727. Correct decisions are 40 + 30 = 70 out of 100, so accuracy is 0.7.

Q4. A GAN is a two-player game: one network forges samples while a critic network tries to catch the forgeries. A diffusion model is trained to undo a gradual corruption of the data by Gaussian noise, i.e. to predict the noise at each step. The two-player game often fails to converge, while the denoising regression loss trains smoothly. Forgers can settle on a handful of outputs that fool the critic (mode collapse); denoisers cover the whole distribution better. On the other hand a GAN produces an image in one network evaluation, while a diffusion model must run its denoiser many times.

Q5. Before any question arrives, the corpus is cut into passages, each passage is mapped to a vector and the vectors are indexed. A question is mapped into the same vector space and its nearest passages are looked up; those passages are given to the language model inside its prompt as the evidence to answer from. Weak point one: lexical mismatch means the true passage is not among the nearest ones - fix with combined keyword + vector search plus a re-ranking model. Weak point two: the model answers from memory instead of the passages - fix by requiring citations to passage ids and refusing when no passage supports the answer.